

Question Paper 2015 March**Class 12 Physics, Paper–III****Time: 3 hours****Maximum Marks: 150**

Instruction: (1) Check the question paper for fairness of printing. If there is any lack of fairness, inform the Hall Supervisor immediately.

(2) Use Black or Blue ink to write and pencil to draw diagrams.

Part–1 **$30 \times 1 = 30$**

**Note: (i) Answer all the questions.
(ii) Choose and write the correct answer.**

1. The self-inductance of a straight conductor is:

Ans: The self-inductance of a straight conductor is zero because of the absence of enclosed loops in the straight conductor.

2. When a material particle of mass m_0 attains the velocity of light, its mass becomes:

Ans: As an object approaches the velocity of light that is 186,000 its mass rises very fast and becomes infinite so does the energy required to move it.

3. When a drop of water is introduced between the glass plate and plano convex lens in newtons rings system, the ring system:

Ans: The ring system contracts because the diameter of the rings will increase as it is inversely proportional to the refractive index.

4. The average energy released per fission is:

Ans: The average energy released per fission is 200 MeV that is fusion means breaking up of a heavier nucleus to form two light nuclei.

5. Which of the following is lepton?

Ans: electron is one of the six leptons.

6. The direction of the electric field in the millikan's oil experiments acts:

Ans: The direction of the electric field in the millikans's oil experiments acts as downwards so that the drop particles rises.

7. In a X-ray tube the intensity of the emitted X-ray beam is increased by:

Ans: In a X-ray tube intensity of the emitted X-ray beam is increased by increasing the target potential as it increases the velocity of the electrons.

8. The nature of the electrostatic force and the nuclear force between the neutron and the proton inside the nucleus are respectively:

Ans: The nature of the electrostatic and nuclear force between neutron and proton is both attractive.

10. In an AC circuit the instantaneous values of emf and current are $e=200 (\omega t - \pi/3)$; $i= (\omega t + \pi/6)$ the phase relation between the current and the voltage is:

Ans: The relation between the current and the voltage is the voltage lags behind current by phase angle of $\pi/3$.

11. Electromagnetic waves are:

Ans: Electromagnetic waves may be transverse or longitudinal in nature.

12. Which of the following quantities are scalar:

Ans: Electric force is a scalar quantity as it doesn't depend on the direction.

13. Point charge $+q$, $+q$, $-q$, $-q$ are placed at the corners A, B, C and D respectively. O is the point of intersection of diagonals AC and BD. The resultant electric field intensity at point O is:

Ans: The resultant electric field intensity is zero as the electric potential is also zero.

14. Phosphor-bronze wire is used in the suspension in a moving coil galvanometer because it has:

Ans: phosphor bronze wire is used in the suspension in a moving coil galvanometer because it has small couple per unit twist.

15. In phase modulation:

Ans: In phase modulation only the phase of the carrier waves vary.

16. A coil of area of cross section 0.5 meter square with 10 turns is in the plane perpendicular to a uniform magnetic field of 0.2 wb/m sq .the flux through the coil is:

Ans: $\text{Flux} = NBA$

Where N = no. of turns

B = magnetic field

A = area of cross section

According to the formula

$$\text{Flux} = 0.5 \times 0.2 \times 10$$

$$= 1 \text{ wb}$$

17. A light of wavelength 4000 Å travelling a distance of 2 micrometer produces a phase change of:

Ans: the phase change of the light wavelength is zero.

18. Which of the following does not allow d.c to pass through:

Ans: Capacitor doesn't allow d.c to pass through.

19. A logic gate which has output 1 only when the complements are input to each other:

Ans: The NOR gate has output 1 only when the complements are input to each other.

20. In joule's calorimeter experiment when a current of 1 ampere is passed through a coil for a known interval of time t the temperature of the water increases from 30 to 33 degree Celsius. When a current of 2 ampere is passed through the same coil placed in the same quantity of water and for the same time, the temperature of the water increases from 30 degree Celsius to :

Ans: The temperature will change from 30 to 36 degree Celsius when a current of 2 ampere is passed through the same coil placed in the same quantity of water and for the same time.

21. A hollow metal ball carrying an electric charge produces no electric field at points:

Ans: A hollow metal ball carrying an electric charge produces no electric field inside the sphere as an electric field inside the sphere is zero.

22. The electrons in the atom of an element which determine its chemical and electrical properties are:

Ans: Valence electrons determine the chemical and electrical properties in the atom of an element.

23. Printed documents to be transmitted by a fax are converted into electrical signals by the process of:

Ans: The printed documents to be transmitted by a fax are converted into electrical signals by scanning.

24. The unit of molecular polarisability is:

Ans: The unit of molecular polarisability is $\text{C}^2\text{m}^2\text{V}^{-1}$.

25. The stopping potential of a metal surface is independent of:

Ans: The stopping potential of a metal surface is independent of the intensity of the incident radiation.

26. In hydrogen atom which of the following transition produces a spectral line of maximum frequency:

Ans: $n=3$ to $n=1$ transition produces a spectral line of maximum frequency.

27. A graph is drawn taking potential difference across the ends of a conductor along the X-axis and current through the conductor along the Y axis .The slope of the straight line gives:

Ans: The slope of the straight line gives the resistance.

28. In beta decay:

In beta decay neutron number decreases by one.

29. The ratio of the radii of the first three bohrs orbit is:

Ans: The ratio of the radii of the first three bohr's orbit is 1:4:9.

30. In a plane transmission grating the width of a ruling is 12000Å and the width of the slit is 8000Å, the grating element is:

Ans: The grating element is 200 micrometer.

PART II

$$15 \times 3 = 45$$

Note: Answer any fifteen questions.

31. State columbs law in electrostatics:

Ans: Columbs law states that the magnitude of the electrostatic force of attraction or repulsion between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them.

32.A sample of HCL gas is placed in an electric field of 2.5×10^4 .the dipole moment of each HCL molecule is 3.4×10^{-30} cm. Find the maximum torque that can act on the molecule.

Ans: $E = 2.5 \times 10^4$

$P = 3.4 \times 10^{-30}$

Torque = $PE \sin \theta$

$$= 3.4 \times 10^{-30} \times 2.5 \times 10^4 \times 1$$

$$= 8.5 \times 10^{-26}$$

33.Define resistivity of a material. Give its S.I unit:

Ans: The electrical resistivity is the electrical resistance which stop the flow of electrons, per unit length and per unit of cross sectional area at a specific temperature. The S.I unit of electrical resistivity is the ohm meter.

34.Compare emf and potential difference:

Ans: Emf is the maximum potential difference between the two electrodes of the cell in an open circuit. It is independent of resistance of the circuit.

Whereas

Potential difference is the difference of potentials between two points in a closed circuit. It is proportional to the resistance of the circuit.

35. Write any three applications of superconductor:

Ans: Superconductors are used in magnetic resonance imaging(MRI),nuclear magnetic resonance(NMR),magnetic confinement fusion reactors.

36. State Flemings left hand rule:

Ans: Fleming's left-hand rule states that if we stretch the thumb, the forefinger and the middle finger of our left hand such that they are mutually perpendicular to each other. If the forefinger gives the direction of current and middle finger points in the direction of magnetic field then the thumb points towards the direction of the force or motion of the conductor.

37. An AC generator consists of a coil of 10,000 turns and of area 100 cm square .The coil rotates at a regular angular speed of 140 rpm in a uniform magnetic field of 3.6×10^{-2} T. find the maximum value of emf induced:

Ans: $N=10,000$

$A=100 \text{ cm square}=10 \text{ power}-2 \text{ meter}$

$v=140 \text{ rpm}=140/60=7/3 \text{ rps}$

$B=3.6 \times 10 \text{ power}-2$

$\text{Emf induced} = NAB (2\pi v)$

$$=10000 \times 1/100 \times 3.6 \times 1/100 \times 2 \times 22/7 \times 7/3$$

$$=52.8 \text{ volt}$$

38. Define quality factor:

Ans: The quality factor is the resonance frequency divided by resonator bandwidth.

39. On what a factor does the amount of optical rotation depends:

Ans: Optical rotation depends on concentration path length temperature wavelength of the light used and the solvent.

40. A plano convex lens of radius 3 cm is placed on a optically flat glass plate and is illuminated by monochromatic light. The radius of the 8th dark ring is 3.6mm. Calculate the wavelength of the light used.

Ans: $R(8) = \sqrt{(nR/\lambda)}$

$$3.6 \times 10^{-4} = \sqrt{8 \times \text{wavelength} \times 3}$$

$$\text{Wavelength} = 5.4 \times 10^{-9} \text{ cm}$$

41. What is ionization potential of an atom:

Ans: The ionization energy or ionization potential is the energy necessary to remove an electron from the neutral atom. It is a minimum for the alkali metals which have a single electron outside a closed shell. It generally increases across a row on the periodic maximum for the noble gases which have closed shells.

42. The rydberg constant for hydrogen atom is 1.097×10^7 per meter. Calculate the long wavelength limit of lyman series.

$$\text{Ans: } 1/\text{wavelength} = 1.097 \times 10^7 (1/1^2 - 1/2^2)$$

$$\text{Wavelength} = 1215.67 \times 10^{-6} \text{ meter}$$

43. Define threshold frequency:

Ans: Threshold frequency is the minimum frequency of light which causes electrons to be emitted from a metal surface.

44. Define curie.

Ans: Curie is the unit of radioactivity corresponding to 3.7×10^{10} disintegrations per second.

45. What are the precautions to be taken by the people who are working in the radiation laboratories:

Ans: The precautions to be taken are use time, distance, shielding, containment to reduce exposure, Wear dosimeters if issued, avoid contact with the contamination, wear protective clothing that if contaminated can be removed.

46. Different methods of doping a semiconductor:

Ans: There are two techniques of doping ion implantation and diffusion.

47. State demorgans theorem:

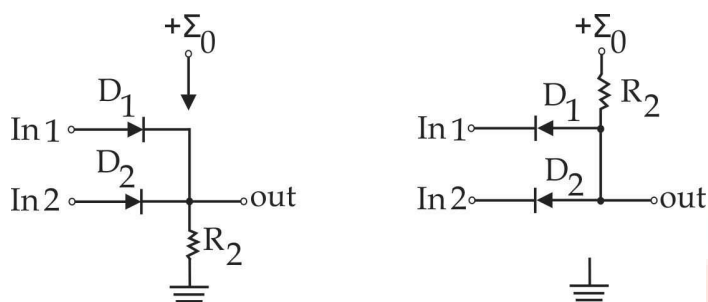
Ans: Demorgans first theorem states that the complement of the sum of the two or more variables is equal to the product of the complement of the variables. The second theorem states that the complement of the product of two or more variables is equal to the sum of the complements of the variables.

48. Draw the circuit diagram of AND gates using diodes.

Ans:

OR

AND



In 1	In 2	Out
0	0	0
0	1	1
1	1	1

0 - OV
1 = Σa

In 1	In 2	Out
0	0	0
0	1	0
1	1	1

50. Define modulation factor in amplitude modulation:

Ans: Modulation factor describes the extent to which the amplitude of the carrier wave is changed by the audio signal:

PART III

$$7 \times 5 = 35$$

Note: (i) Answer Question No.53 compulsorily.

(ii) Answer any six of the remaining 11 questions.

(iii) Draw diagrams wherever necessary.

51. Write the properties of electric lines of forces:

Ans: The properties lines of force start from positive charge and ends on a negative charge. The electric lines of forces always enter or leave the charged surface normally. Electric lines of force can never intersect each other. The electric lines of force

cannot pass through a conductor. When two like charges are placed closer to each other the electric lines of force present between them will become enlarged in length.

52. Obtain the condition of the bridge balance in the wheatstone bridge:

Ans: The ratio of the arms R_1 and R_2 is equal to the ratio of the arms R_3 and R_4 this is the balanced condition in the wheatstone bridge.

53. The effective resistance are 10 ohm, 2.4 ohm when two resistors are connected in series and parallel .What are the resistances of individual resistors:

Ans: $R_1 + R_2 = 10$ when in series

$$1/R_1 + 1/R_2 = 1/2.4 \text{ when in parallel}$$

$$(R_1 \times R_2)/(R_1 + R_2) = 2.4$$

$$(R_1 \times R_2) = 2.4 \times 10$$

$$(R_1 \times R_2) = 24$$

$$R_1 + 24/R_1 = 10 \text{ from first equation}$$

On solving we get

$$R_1 = 4 \text{ or } 6 \text{ ohm}$$

$$R_2 = 6 \text{ ohm and vice versa}$$

54. A moving coil of galvanometer of resistance 20 ohm produces full scale deflection for a current of 50mA. How will you convert the galvanometer into:

A) An ammeter of range 20 A

B) A voltmeter of range 120 v

Ans: converting to ammeter

$$50\text{mA} = 0.05\text{A}$$

$$20\text{ ohm}$$

$$20 \times 0.05 = 1\text{V}$$

$$\text{Current across shunt} = 20 - 0.05 = 19.95\text{A}$$

$$20 \times 0.05 = S \times 19.95$$

$$S = 1/19.95$$

$$S = 0.05\text{ ohm (parallel)}$$

Converting to voltmeter

High resistance must have a drop of $120 - 1 = 119\text{V}$ when 0.05A current flows through it.

$$0.05R = 119$$

$$R = 2380\text{ ohm (series)}$$

55. Explain the energy losses in a transformer .how are they minimized:

Ans: The energy losses in a transformer are of two types: hysteresis loss and eddy current loss .The minimize the losses core made up of silicon steel or crgo steel is used or by staggering the iron core.

56. State and prove Brewster's law:

Ans: Brewster's law is the relationship for the light waves stating that the maximum polarization of a ray of light may be achieved by letting the ray fall on the surface of a transparent medium in such way that the refracted ray makes an angle of 90 degree with the reflected ray.

$$u = \tan i_p$$

$$u = (\sin i_p) / (\sin i_r)$$

$$\tan i_p = (\sin i_p) / (\sin i_r)$$

57. Write any five properties of X-ray:

Ans: They have wavelength between 10Å to 0.01Å, in free space they travel in straight line, it cannot be focused on a single point, they are invisible to the eye, they cannot be heard.

58. Explain Lorentz–fitzgerald contraction with example:

Ans: The shortening of an object along its direction of motion as its speed approaches the speed of light, as measured by an observer at rest with respect to the object. For example consider a train carriage that moves with constant speed along a straight track, past us who stand on the platform watching it. Light that leaves the back of the carriage will arrive at our eyes coincident with light from the side of the carriage as well as the carriage is Lorentz contracted.

59. Derive an expressive for the de Broglie wavelength of matter waves.

Ans: $E = mc^2$

$$E = hc \lambda$$

$$\text{Wavelength} = \frac{h}{mv}$$

60. The binding energy per nuclear for carbon 12 nucleus is 7.68MeV and that for carbon 13 is 7.47MeV. calculate energy required to move electron from carbon 13:

Ans: Total binding energy of carbon 12 = $12 \times 7.68 = 92.16 \text{ MeV}$

Total binding energy of carbon 13 = $13 \times 7.47 = 97.11 \text{ MeV}$

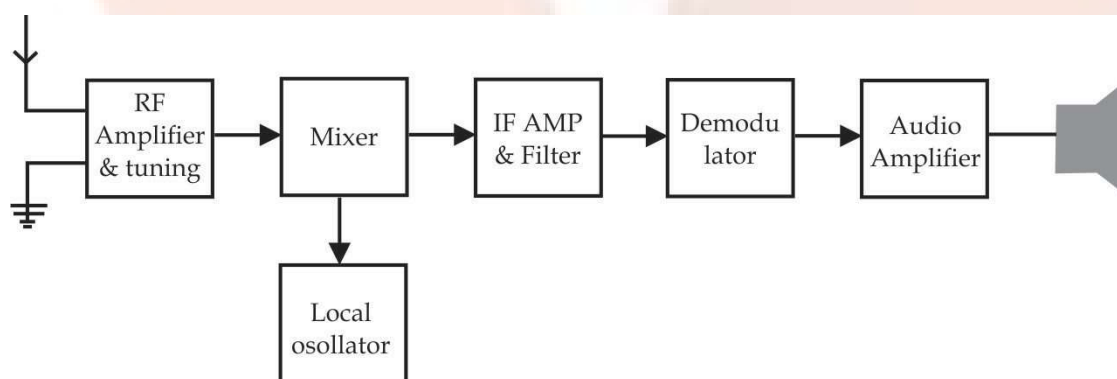
Energy required to move an electron = $97.11 - 92.16 = 4.95 \text{ MeV}$

61. Explain function of transistor as switch:

Ans: in transistor, unless a current flows in the base circuit there is current can flow in the collector circuit. The transistor can be switched on and off by changing the base. There are a few applications of switching circuits operated by transistors.

62. With the help of a block diagram explain the operation of FM super heterodyne receiver:

Ans: the block diagram explains that the rf amplifies the received signal intercepted by the antenna. The amplified signal is then applied to the mixer stage. The second input of the mixer comes from the local oscillator. The two frequencies of mixer generate an IF signal of 10.7 MHz. This signal is then amplified by the IF amplifier. The output of the IF amplifier is applied to the limiter circuit. The limiter removes the noise in the received signal and gives a constant amplitude signal. The output of the limiter is now applied to the FM discriminator which recovers the modulating signal.



PART IV

$$4 \times 10 = 40$$

Note: (i) Answer any four questions.

(ii) Draw diagrams wherever necessary.

63. Obtain an expression for electric potential at a point due to an electric dipole .discuss the special cases:

$$\text{Ans: } E(q) = kq/(r-a) - kq/(r+a) = kq(2a/(r^2 - a^2))$$

64 .Obtain a expression for the magnetic induction at a point due to infinitely long straight conductor carrying current:

$$\text{Ans: } B = \mu_0/4\pi \times I dl \sin\theta / r^2$$

$$\text{In ABC triangle } \sin\theta = AC/AB = AC/dl$$

$$AC = r \sin\theta$$

$$B = \mu_0/4\pi \times I/a \times \cos\theta_1 - \cos\theta_2$$

By integrating

$$B = \mu_0 I / 4\pi a (\sin\theta_1 + \sin\theta_2)$$

65. Discuss with the theory the method of inducing emf in a coil by changing its orientation with respect to direction of magnetic field:

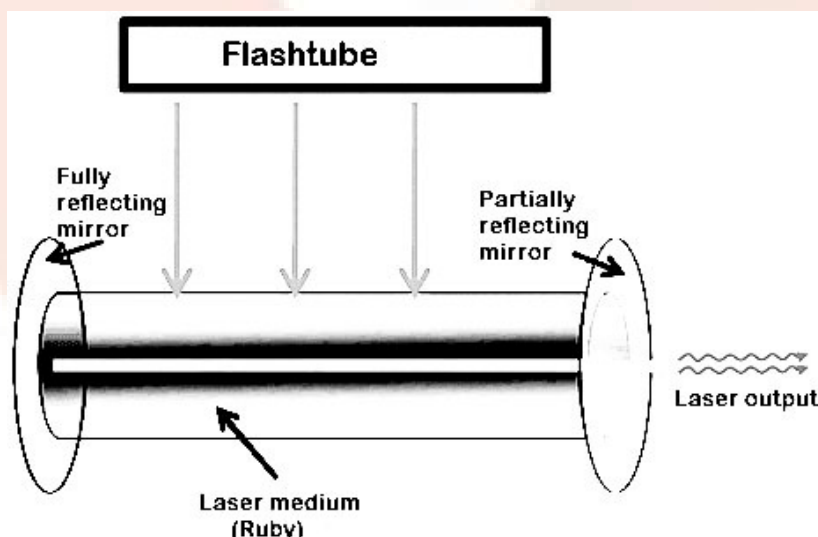
Ans : By increasing the strength of the magnetic field – If the same coil of wire is moved at the same speed through a stronger magnetic field, there will be more emf produced because there are more lines of force to cut.

66. What is emission and absorption spectrum. Explain the different types of emission and absorption spectrum with examples:

Ans: If a large enough voltage difference is applied between the two metal plates, the gas atoms inside the discharge tube will absorb enough energy to make some of their electrons come off, i.e. the gas atoms are ionized. These electrons start moving through the gas and create a current, which raises some electrons in other atoms to higher energy levels. Then as the electrons in the atoms fall back down, they emit electromagnetic radiation (light). The amount of light emitted at different wavelengths, called the emission spectrum. Atoms do not only emit photons; they also absorb photons. If a photon hits an atom and the energy of the photon is the same as the gap between two electron energy levels in the atom, then the electron in the lower energy level can absorb the photon and jump up to the higher energy level. The absorbed photons show up as black lines because the photons of these wavelengths have been absorbed and do not show up. Because of this, the absorption spectrum is the exact inverse of the emission spectrum.

67. Draw a neat sketch of ruby laser. Explain its working with the help of energy level diagram:

Ans: When light energy is supplied to the laser medium (ruby), the electrons in the lower energy state or ground state (E_1) gain energy and jump into the higher (pump) state (E_3). The lifetime of pump state E_3 is very small (10^{-8} sec) so the electrons in the pump state do not stay for long period. After a short period, electrons fall into the stable state E_2 by releasing radiation as energy. The lifetime of metastable state E_2 is 10^{-3} sec which is much greater than the lifetime of pump state E_3 . Therefore, the electrons reach E_2 much faster than they leave E_2 . Thus an increase in the number of electrons in the stable state E_2 and hence population inversion is achieved. The sketch of ruby laser is given below.



68: What are cosmic rays. Explain:

A) Latitude effect

B) Altitude effect of cosmic rays:

Ans: Ionizing radiation coming from the outer space is the reason for leakage of charges in the gold leaf electroscope. The ionizing radiation many times stronger than γ -rays entering the earth from all the directions from cosmic or interstellar space is known as cosmic rays. The variation of cosmic ray intensity (I) with geomagnetic latitude (θ) showed that the intensity is maximum at the poles ($\theta = 90^\circ$), minimum at the equator ($\theta = 0^\circ$) and constant between latitudes of 42° and 90° . The variation of cosmic ray intensity with geomagnetic latitude is known as latitude effect. Variation of cosmic ray intensity (I) with altitude (h) is known as altitude effect,

69. What is meant by feedback. Derive an expression for voltage gain of an amplifier with negative feedback:

Ans: Feedback is the process by which a fraction of the output signal, either a voltage or a current, is used as an input. If this feedback fraction is opposite in value or phase to the input signal, then the feedback is said to be Negative Feedback.

Expression—system gain= $G = V_{out}/V_{in}$

$$G \times V_{in} = V_{out}$$

$$G(V_{in} - BV_{out}) = V_{out}$$

$$G V_{in} - BG V_{out} = V_{out}$$

$$G V_{in} = V_{out} (1 + BG)$$

$$V_{out}/V_{in} = G_v = G/1 + BG$$

70. Explain transmission and reception of radar with block diagram:

Ans: Transmission and Reception of Radar: The block diagram of a simple radar system is shown in Figure. This block diagram indicates that the radar system consists of both the transmitting and the receiving system.

The transmitting system consists of a transmitter and a pulser. The receiving system consists of a receiver and an indicator. In most of the cases, a single antenna is used for both transmission and reception and this is achieved with the use of TR switch (Transmitter Receiver Switch). This connects the antenna to the transmitter during transmission and to the receiver during reception. Moreover, this switch isolates the sensitive receiver from the damaging effects of the high power transmitter–receiver. The sensitivity of the receiver is very high. The detected output is sent to the indicator. The indicator is a cathode ray tube. The CRT displays the original transmitted

pulse as well as the detected echo pulse along a horizontal base line. The synchronizing pulse generated by the 'timer' is supplied to both transmitting and receiving systems. So, the indicator records the transmitted pulse as well as the returning pulse simultaneously. The returning echo pulse appears slightly displaced from the transmitted pulse and this displacement is a measure of the range of the target.

The block diagram is drawn below.

